CRD_Analylsis

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Completely Randomized Design (CRD)

A **Completely Randomized Design (CRD)** is used when experimental units are reasonably homogeneous and treatments can be assigned completely at random.

In this example, different fertilizer treatments are compared based on crop yield.

The experimental structure is:

- **Experimental unit:** plot

- **Treatment:** fertilizer treatment
- **Response variable:** yield (t/ha)
- **Replications:** 10 plots per treatment
- **Design:** CRD
- **Block:** None

The main question is:

> **Do the fertilizer treatments produce different mean yields?**

---

1. Data

Load the data

```
datacrd <- read.csv("crd_normal_data.csv")

head(datacrd)
```

```
  treatment yield_t_ha
1   Control       3.42
2   Control       3.51
3   Control       3.38
4   Control       3.47
5   Control       3.55
6   Control       3.44
```

Check whether there are missing values:

```
colSums(is.na(datacrd))
```

```
 treatment yield_t_ha
         0          0
```

---

2. Model Building

Because this is a CRD, the basic one-way ANOVA model contains only the treatment effect:

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

Fit the ANOVA model

```
modelcrd <- aov(
  yield_t_ha ~ treatment,
  data = datacrd
)

modelcrd
```

```
Call:
   aov(formula = yield_t_ha ~ treatment, data = datacrd)

Terms:
                treatment Residuals
Sum of Squares   6.911793  0.228180
Deg. of Freedom         5        54

Residual standard error: 0.06500427
Estimated effects may be unbalanced
```

View the ANOVA table:

```
summary(modelcrd)
```

```
            Df Sum Sq Mean Sq F value Pr(>F)
treatment    5  6.912  1.3824   327.1 <2e-16 ***
Residuals   54  0.228  0.0042
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

At this point, we have **fitted the model**, but we should not interpret the ANOVA p-value yet.

First, we need to check whether the assumptions of the model are reasonably satisfied.

3. Model Diagnostics

Before interpreting the ANOVA result, we need to check whether the model assumptions are reasonably satisfied.

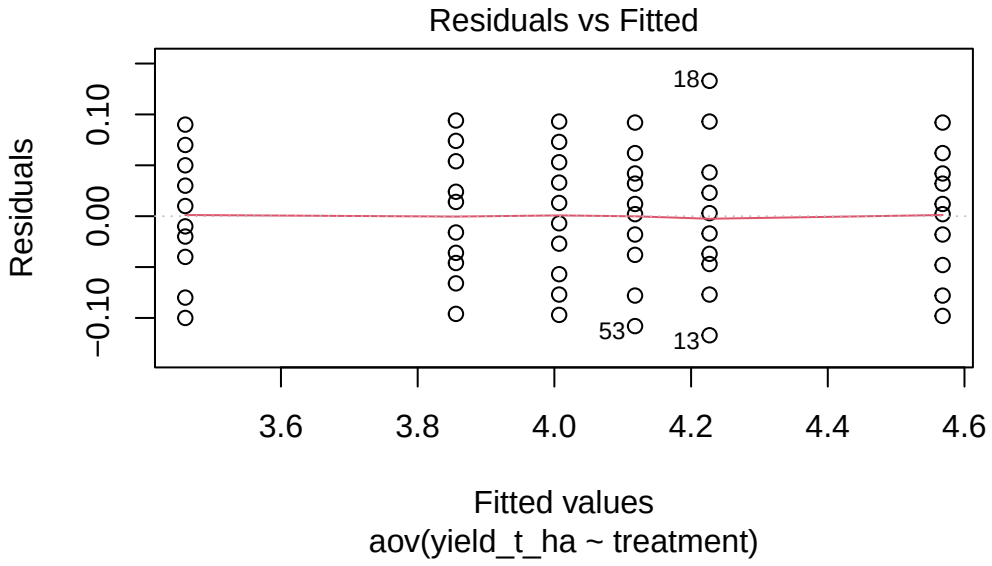
For this model, the main assumptions are:

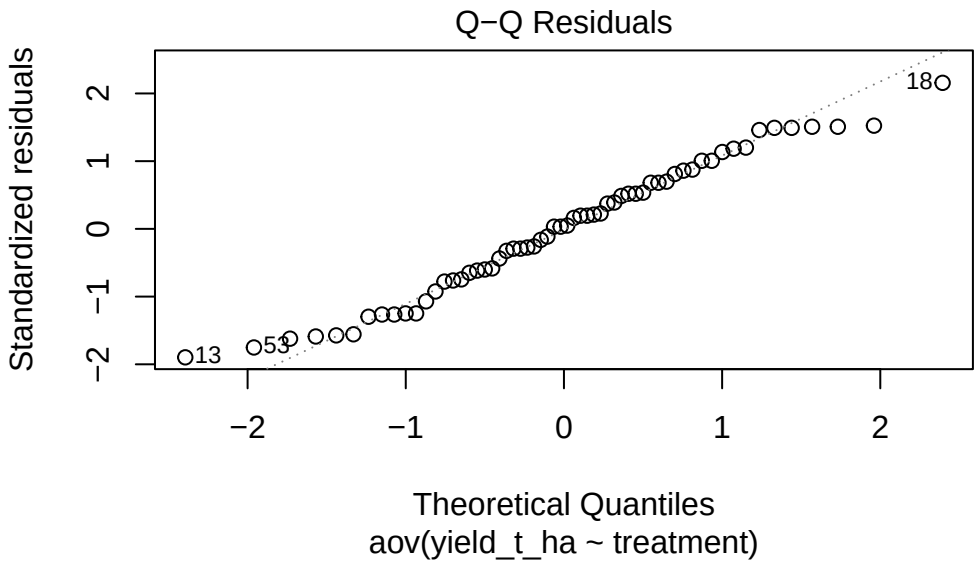
1. Independence of observations
2. Approximately normal residuals
3. Homogeneity of residual variance

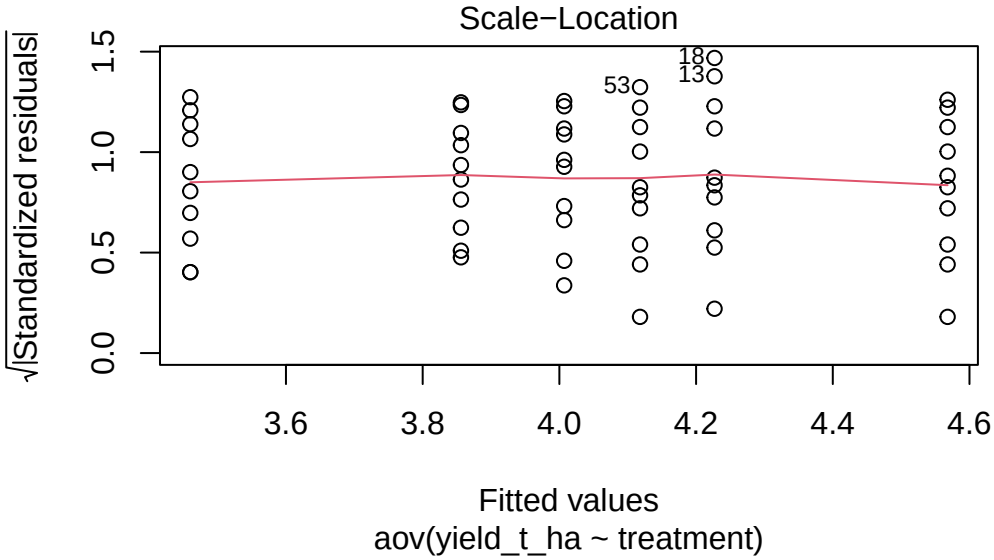
3.1 Diagnostic plots

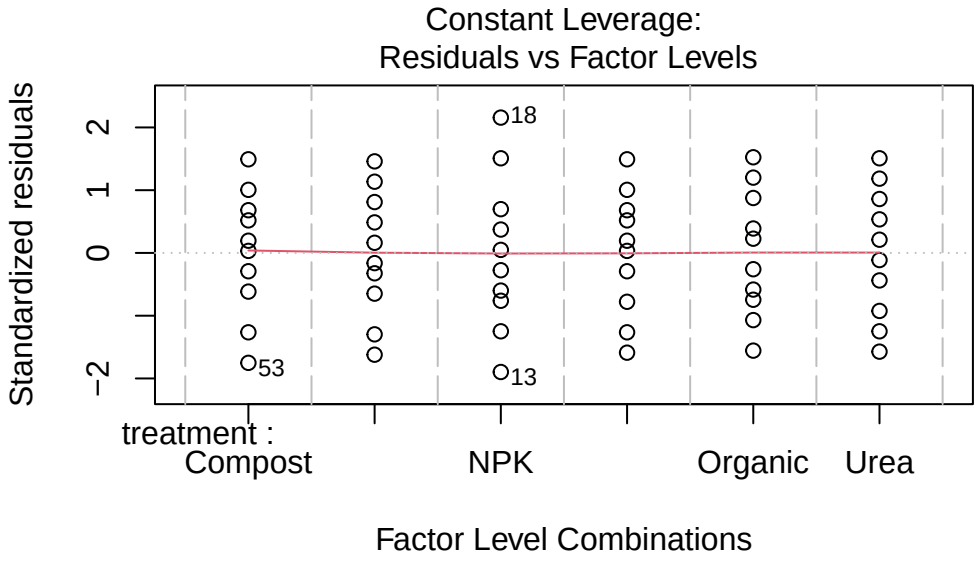
The standard diagnostic plots can be obtained from the fitted model.

```
plot(modelcrd)
```




Q−Q Residuals
Standardized residuals
−2
−1
0
1
2
18
13
53
Theoretical Quantiles
aov(yield_t_ha ~ treatment)


Scale−Location
√|Standardized residuals|
0.0
0.5
1.0
1.5
18
13
53
3.6
3.8
4.0
4.2
4.4
4.6
Fitted values
aov(yield_t_ha ~ treatment)



The most important plots for this analysis are:

- **Residuals vs Fitted:** checks whether residual spread is reasonably constant and whether there is a systematic pattern.Look for the random spread(no clear pattern)
- **Normal Q-Q:** checks whether residuals approximately follow a normal distribution. If the points approximately follow the reference line, the residual distribution is reasonably close to normal.
- **Scale-Location:** helps assess homogeneity of variance.
- **Residuals vs Leverage:** helps identify observations that may have unusually strong influence.

3.2 Normality of residuals

The Q-Q plot is the primary visual check. If the points approximately follow the reference line, the residual distribution is reasonably close to normal.

A formal Shapiro-Wilk test can also be used:

```
shapiro.test(residuals(modelcrd))
```

```
    Shapiro-Wilk normality test
```

```
data:  residuals(modelcrd)
W = 0.97445, p-value = 0.2396
```

Interpretation:

- p value is more than 0.05 the residuals are normally distributed, can perform the one way ANOVA.
- p value is less than 0.05 the residuals are not normally distributed, performing one way ANOVA isn't reliable.

The Shapiro-Wilk test should not be used alone. With larger samples, small departures from normality can become statistically significant even when the ANOVA is reasonably robust.

Therefore, the **Q-Q plot and the practical size of the deviation** should also be considered.

3.3 Homogeneity of variance

A useful formal test is Levene's test.

```
library(car)
```

```
Warning: package 'car' was built under R version 4.4.3
```

```
Loading required package: carData
```

```
leveneTest(yield_t_ha ~ treatment, data = datacrd)
```

```
Warning in leveneTest.default(y = y, group = group, ...): group coerced to
factor.
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  5  0.1341 0.9838
      54
```

Interpretation:

- **$p > 0.05$:** no strong evidence of unequal variances.
- **$p < 0.05$:** evidence that the treatment variances differ.

Again, the test should be considered together with the diagnostic plots.

3.4 Diagnostic decision

For this dataset, the residual diagnostics indicate that the assumptions are reasonably satisfied.

Therefore:

> **Decision:** Proceed with the one-way ANOVA model.

If the assumptions were seriously violated, we would reconsider the analysis rather than blindly interpreting the ANOVA result. For a CRD with a non-normal response, a non-parametric alternative such as the **Kruskal-Wallis test** may be considered.

---

4. ANOVA Interpretation

Once the model assumptions are reasonably satisfied, we can interpret the ANOVA table.

```
summary(modelcrd)
```

```
            Df Sum Sq Mean Sq F value Pr(>F)
treatment    5  6.912  1.3824   327.1 <2e-16 ***
Residuals   54  0.228  0.0042
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The hypotheses are:

- **Null hypothesis (H):** all treatment means are equal.
- **Alternative hypothesis (H):** at least one treatment mean differs from another.

The ANOVA table provides an **F-statistic** and a **p-value** for the treatment effect.

Statistical decision

Use the significance level:

$$\alpha = 0.05$$

Decision rule:

- **p $<$ 0.05:** reject H .
- **p 0.05:** fail to reject H .

For this dataset, the treatment effect is statistically significant.

Therefore:

> **Decision:** Reject H . There is evidence that mean yield is not the same across all fertilizer treatments.

However, ANOVA does **not** tell us which specific treatments differ.

It only tells us that there is evidence of a difference somewhere among the treatment means.

Therefore, a post-hoc comparison is required.

---

5. Post-hoc Analysis

Post-hoc analysis is performed when the overall ANOVA indicates a statistically significant treatment effect and we want to determine **which treatment means differ**.

Here we use `emmeans` to obtain estimated marginal means and compare treatments.

5.1 LSD comparison

For an LSD-style comparison, we can use no multiplicity adjustment:

```
library(emmeans)
```

```
Warning: package 'emmeans' was built under R version 4.4.3
```

```
Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'
```

```r
emm_crd_lsd = emmeans(modelcrd,pairwise~treatment, adjust="none")
emm_crd_lsd
```

```
$emmeans
 treatment   emmean     SE df lower.CL upper.CL
 Compost       4.12 0.0206 54     4.08     4.16
 Control       3.46 0.0206 54     3.42     3.50
 NPK           4.23 0.0206 54     4.19     4.27
 NPK_Organic   4.57 0.0206 54     4.53     4.61
 Organic       3.86 0.0206 54     3.81     3.90
 Urea          4.01 0.0206 54     3.97     4.05

Confidence level used: 0.95

$contrasts
 contrast              estimate     SE df t.ratio p.value
 Compost - Control        0.658 0.0291 54  22.634  <.0001
 Compost - NPK           -0.109 0.0291 54  -3.749  0.0004
 Compost - NPK_Organic   -0.450 0.0291 54 -15.479  <.0001
 Compost - Organic        0.262 0.0291 54   9.012  <.0001
 Compost - Urea           0.111 0.0291 54   3.818  0.0003
 Control - NPK           -0.767 0.0291 54 -26.384  <.0001
 Control - NPK_Organic   -1.108 0.0291 54 -38.114  <.0001
 Control - Organic       -0.396 0.0291 54 -13.622  <.0001
 Control - Urea          -0.547 0.0291 54 -18.816  <.0001
 NPK - NPK_Organic       -0.341 0.0291 54 -11.730  <.0001
 NPK - Organic            0.371 0.0291 54  12.762  <.0001
 NPK - Urea               0.220 0.0291 54   7.568  <.0001
 NPK_Organic - Organic    0.712 0.0291 54  24.492  <.0001
 NPK_Organic - Urea       0.561 0.0291 54  19.298  <.0001
 Organic - Urea          -0.151 0.0291 54  -5.194  <.0001
```

A compact letter display can be produced using `multcomp`:

```r
library(multcomp)
```

```
Warning: package 'multcomp' was built under R version 4.4.3
```

```
Loading required package: mvtnorm

Loading required package: survival

Loading required package: TH.data

Loading required package: MASS


Attaching package: 'TH.data'

The following object is masked from 'package:MASS':

    geyser
```

```
library(multcompView)
```

```
Warning: package 'multcompView' was built under R version 4.4.3
```

```
multcomp::cld( emm_crd_lsd, adjust = "none", Letters = letters )
```

```
 treatment   emmean     SE df lower.CL upper.CL .group
 Control       3.46 0.0206 54     3.42     3.50  a
 Organic       3.86 0.0206 54     3.81     3.90   b
 Urea          4.01 0.0206 54     3.97     4.05    c
 Compost       4.12 0.0206 54     4.08     4.16     d
 NPK           4.23 0.0206 54     4.19     4.27      e
 NPK_Organic   4.57 0.0206 54     4.53     4.61       f

Confidence level used: 0.95
significance level used: alpha = 0.05
NOTE: If two or more means share the same grouping symbol,
      then we cannot show them to be different.
      But we also did not show them to be the same.
```

Treatments sharing the same letter are **not statistically different** under the selected comparison procedure.

Treatments with different letters are statistically different.

5.2 Tukey HSD comparison

Tukey's method controls the family-wise error rate when making multiple pairwise comparisons.

```
library(emmeans)
emm_crd_hsd = emmeans(modelcrd,pairwise~treatment, adjust="tukey")
emm_crd_hsd
```

```
$emmeans
 treatment   emmean     SE df lower.CL upper.CL
 Compost       4.12 0.0206 54     4.08     4.16
 Control       3.46 0.0206 54     3.42     3.50
 NPK           4.23 0.0206 54     4.19     4.27
 NPK_Organic   4.57 0.0206 54     4.53     4.61
 Organic       3.86 0.0206 54     3.81     3.90
 Urea          4.01 0.0206 54     3.97     4.05

Confidence level used: 0.95

$contrasts
 contrast              estimate     SE df t.ratio p.value
 Compost - Control        0.658 0.0291 54  22.634  <.0001
 Compost - NPK           -0.109 0.0291 54  -3.749  0.0055
 Compost - NPK_Organic   -0.450 0.0291 54 -15.479  <.0001
 Compost - Organic        0.262 0.0291 54   9.012  <.0001
 Compost - Urea           0.111 0.0291 54   3.818  0.0045
 Control - NPK           -0.767 0.0291 54 -26.384  <.0001
 Control - NPK_Organic   -1.108 0.0291 54 -38.114  <.0001
 Control - Organic       -0.396 0.0291 54 -13.622  <.0001
 Control - Urea          -0.547 0.0291 54 -18.816  <.0001
 NPK - NPK_Organic       -0.341 0.0291 54 -11.730  <.0001
 NPK - Organic            0.371 0.0291 54  12.762  <.0001
 NPK - Urea               0.220 0.0291 54   7.568  <.0001
 NPK_Organic - Organic    0.712 0.0291 54  24.492  <.0001
 NPK_Organic - Urea       0.561 0.0291 54  19.298  <.0001
 Organic - Urea          -0.151 0.0291 54  -5.194  <.0001

P value adjustment: tukey method for comparing a family of 6 estimates
```

Compact letter display:

```r
multcomp::cld( emm_crd_hsd, adjust = "sidak", Letters = letters )
```

```
 treatment   emmean     SE df lower.CL upper.CL .group
 Control       3.46 0.0206 54     3.40     3.52  a
 Organic       3.86 0.0206 54     3.80     3.91   b
 Urea          4.01 0.0206 54     3.95     4.06    c
 Compost       4.12 0.0206 54     4.06     4.17     d
 NPK           4.23 0.0206 54     4.17     4.28      e
 NPK_Organic   4.57 0.0206 54     4.51     4.62       f

Confidence level used: 0.95
Conf-level adjustment: sidak method for 6 estimates
P value adjustment: sidak method for 15 tests
significance level used: alpha = 0.05
NOTE: If two or more means share the same grouping symbol,
      then we cannot show them to be different.
      But we also did not show them to be the same.
```

LSD vs Tukey

The two approaches answer the same general question—**which treatment means differ?**—but they control multiple comparisons differently.

- **LSD:** less conservative; greater chance of detecting differences but greater risk of Type I error across many comparisons.
- **Tukey:** more conservative and appropriate when making all pairwise treatment comparisons.

For a general-purpose research analysis, **Tukey-adjusted comparisons are usually the safer default** when all treatment pairs are being compared.

The post-hoc results will be used in the next section to determine **which treatments differ and how those differences should be interpreted agriculturally**.

6. Post-hoc Interpretation

Tukey's test shows that **all treatment pairs are significantly different** at $= 0.05$.

The treatment ranking based on estimated mean yield is:

```
NPK_Organic > NPK > Compost > Urea > Organic > Control
```

The compact letter display confirms this because each treatment has a different letter.

Therefore, `NPK_Organic` produced the highest yield and was significantly higher than all other treatments.

---

7. Agricultural Interpretation

`NPK_Organic` produced the highest estimated yield (**4.57 t/ha**), while the Control produced the lowest (**3.46 t/ha**).

The difference was approximately **1.11 t/ha** and was statistically significant.

However, statistical significance alone does not mean that `NPK_Organic` is the best practical recommendation. Cost, fertilizer rate, economic return, and environmental effects should also be considered.